

Spatial Intelligence for Urban Greening: Identifying Priority Locations and Greening Interventions

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Abstract

The aim of the project would be to develop a tool or model, that would identify suitable locations on campus for greenery integration – both tree planting and Vertical Greenery Systems.

Introduction:

Greenery is an integral part of Singapore's urban landscape. City in Nature, a SG Green Plan 2030 pillar, aims to achieve its 1 million trees planted target by 2030 (SG Green Plan 2030, 2026). NParks also echoes a similar commitment to have 200 ha of skyrise greenery and 170,000 trees planted in industrial areas by 2030 (National Parks Board, 2026).

Similar campus-wide tree planting efforts have also been taken at NUS campus since 2020. However, tree planting was largely performed on green areas. Examples of VGS integration at the non-green areas on campus include Ventus and SDE 4. Hence, non-green areas could be better utilized for VGS integration.

Green roofs are also more widely adopted than green walls for VGS integration.

Hence, the tool would be designed to locate suitable green and non-green areas for greenery integration on campus.

Related Work:

Chen et al. (2026) examined the combined effects of urban morphology and greenery on outdoor thermal comfort in Singapore. Their results showed that tree cover was an important factor in thermal comfort prediction, while its cooling effect varied with surrounding building morphology. This suggests that greenery should not be considered independently from the built environment when assessing urban thermal conditions.

Building on this, our project investigates whether adding spatially quantified greenery information can improve pedestrian-level air-temperature prediction at the NUS campus, beyond meteorological and built-morphology variables.

Methods

We would be considering both the following dataset and tool for this study,

1. Dataset: NUS BEAM for extraction to extract climatic variables, NUS Campus Weather (YQ,2025)
2. STEVE Tool: to extract location-specific climatic and/or greenery variables

The following outlines the steps to be taken following interim submission,

1. Further, Greenery related EDA analysis to be performed.
2. A separate Greenery variable would need to be subsequently characterized
3. Subsequently, Random Forest and XG Boost will be used to relate the surroundings of each weather station (tree canopy, building coverage, elevation) to its temperature anomaly relative to the campus mean, with solar radiation, humidity and wind as control variables. These two tree-based models are good at modelling non-linear relationships.
4. A linear regression would serve as a baseline.
5. Models will be tested on stations left out of training. Feature importance will then show which variables matter most.

6. Priority locations will be then consequently identified.

The following would be a sample of how the user could use the tool:

1. User to place a tree on the model using a + Tree option
2. The tool will ask the user to key in or select greenery parameters.

Examples of these parameters are as follows:

- Tree Height: 8m
- Canopy: 6m
- Leaf Area Index (LAI): 2.5
- Tree Canopy Coverage: +10%

References:

Singapore Green Plan 2030. (2026, March 31). Our targets. <https://www.greenplan.gov.sg/targets/>

National Parks Board. (2026, September 2). City in Nature: Key strategies. <https://www.nparks.gov.sg/who-we-are/city-in-nature-key-strategies>

Chen, T., Liu, X., & Yuan, C. (2026). Machine learning-based design optimization of urban greening for cooling in tropical cities. *Building and Environment*, 295, 114468. <https://doi.org/10.1016/j.buildenv.2026.114468>